Year 13 Further Maths - Complex numbers and trigonometry

Similar questions - Expressing powers of sin theta and cos theta

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1

(a) By writing $\cos\theta$ in terms of $e^{i\theta}$ and $e^{-i\theta}$, show that $\cos^4\theta = \frac{1}{8}(3+4\cos 2\theta+\cos 4\theta)$. **[5]**

(b) Hence find the exact value of $\int_0^{\pi/2} \cos^4\theta \, d\theta$. **[2]**

2 Use de Moivre's theorem to find the constants A, B and C in the identity $\sin^5\theta \equiv A\sin\theta + B\sin 3\theta + C\sin 5\theta$. **[5]**

Question 1

(a) $\cos\theta = \frac{1}{2}(z+z^{-1})$ with $z = e^{i\theta}$. $\cos^4\theta = \frac{1}{16}(z+z^{-1})^4 = \frac{1}{16}\left[z^4+4z^2+6+4z^{-2}+z^{-4}\right] = \frac{1}{16}\left[(z^4+z^{-4})+4(z^2+z^{-2})+6\right] = \frac{1}{16}\left[2\cos 4\theta + 8\cos 2\theta + 6\right] = \frac{1}{8}(3+4\cos 2\theta+\cos 4\theta)$

(b) $\displaystyle\int_0^{\pi/2} \tfrac{1}{8}(3+4\cos 2\theta+\cos 4\theta)\,d\theta = \tfrac{1}{8}\left[3\theta+2\sin 2\theta+\tfrac{1}{4}\sin 4\theta\right]_0^{\pi/2} = \tfrac{1}{8}\cdot\tfrac{3\pi}{2} = \dfrac{3\pi}{16}$

Question 2

$\sin\theta = \dfrac{z-z^{-1}}{2i}$, so $\sin^5\theta = \dfrac{(z-z^{-1})^5}{32i} = \dfrac{(z^5-z^{-5})-5(z^3-z^{-3})+10(z-z^{-1})}{32i}$. Using $z^n - z^{-n} = 2i\sin n\theta$: $\sin^5\theta = \dfrac{1}{16}(\sin 5\theta - 5\sin 3\theta + 10\sin\theta)$. So $A = \frac{5}{8}$, $B = -\frac{5}{16}$, $C = \frac{1}{16}$